

Technology Stack

Date	2 July 2026
Team ID	SWTID-2026-7147
Project Name	Smart Lender
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, Bootstrap	Creates a responsive and user-friendly web interface for entering loan applicant details and viewing prediction results.
2	Backend / Server-Side	Python, Flask	Flask handles user requests, processes applicant data, and integrates the trained XGBoost model for real-time prediction.
3	Database / Data Storage	CSV Dataset (Training Data)	Stores historical loan applicant data used for training and evaluating the machine learning models.
4	Cloud / Hosting / Deployment	IBM Cloud	Deploys the Flask application online with scalable and reliable cloud hosting.

5	Version Control & CI/CD	GitHub	Maintains source code, enables team collaboration, and tracks project versions.
6	Third-Party APIs / Other Tools	Scikit-learn, XGBoost, Pandas, NumPy, Joblib	Scikit-learn for preprocessing and evaluation, XGBoost for loan prediction, Pandas and NumPy for data processing, and Joblib to save and load the trained model.